

# Detecting forward leaning posture using eSense and developing a posture improvement promoting system

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## ABSTRACT

In this study, we propose a method using eSense for detecting forward leaning posture during computer work, and verify its effectiveness. We will also develop a system that prompts the user to improve posture when a forward leaning posture is detected, and verify its effectiveness. In the future, we aim to develop a system that targets not only forward leaning posture but also posture improvement in other parts of the body by utilizing the contents verified in this research.

## CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

## KEYWORDS

earable device, wearable device, posture improvement, office work

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## 1 INTRODUCTION

Office workers, who work with a computer, often suffer from unnatural posture while working. Working for a long time with unnatural posture can cause not only physical fatigue, but also diseases such as tendinitis and hunchback. In order to prevent the onset of these diseases, it is necessary to work with correct posture continuously. However, it seems difficult to work while always being aware of maintaining correct posture. We think it is necessary that a system that detects unnatural posture and notify users to encourage them

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to maintain correct posture. In this study, we focus on detecting forward leaning posture, and aim to verify the method of it using eSense, which is an earable device equipped with a 6-axis acceleration sensor. At the same time, we aim to develop a system that encourages posture improvement when a forward leaning posture is detected, and to verify the effectiveness of the system.

## 2 RELATED WORK

Several posture estimation methods using wearable devices have been proposed in previous studies. For example, this study[2] verified whether posture estimation is possible using only the 3-axis accelerometer of a cell phone, and found that the basic motions of sitting, standing, walking, and running could be estimated. However, in order to detect forward leaning posture using a cell phone, it needs to be worn on the upper body where body movements are large. Since cell phones have been getting larger and larger on recent years, wearing them on the upper body all the time can be a hindrance to your work. Also, not all people wear a cell phone while working with a computer (for example, some people put it on their desk while working).

On the other hand, there is also previous study about a system that promote improved posture during desk work. For example, this study[3] investigated the use of a very slow moving monitor for unobtrusive posture correction, and revealed its effectiveness. We think this system very good in that it can correct posture imperceptibly without disturbing the user's work with sound or vibration, but the drawback is that it requires a motorized monitor which is hardly widespread yet.

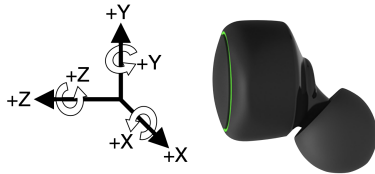
The device used for posture estimation while working on a computer should not interfere with the work and should not cause discomfort even if it is worn for a long time. Therefore, we think earphones are appropriate because it is small and familiar to many people. Although sensor-equipped earphones are not yet widely available, given the popularity of TWS over the past few years, we think it would be useful to conduct this study using earables.

## 3 METHODS

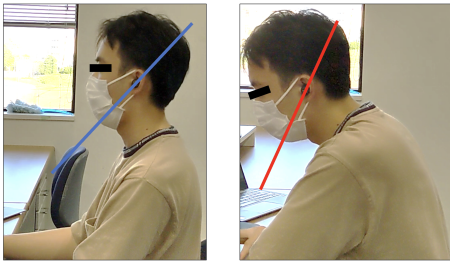
### 3.1 Detecting forward leaning posture

In this study, we use eSense, which is an earable device developed by Nokia Bell Labs[1]. eSense is equipped with a 6-axis accelerometer

in the left earbud, and we propose a method to detect forward-leaning posture from these sensor data (see Fig. 1). To determine whether the posture is forward-leaning or not, we can use the rotation angle of the left earbud. As shown in Figure 2, when users are in forward-leaning posture, there should be a change in the rotation angle on the Z-axis. Therefore, we detect forward-leaning posture when the a rotation angle on the Z-axis of the left earbud exceeds a threshold value for a certain amount of time.



**Figure 1: The orientation of the IMU axes and the polarity of rotation in relation to the earbud (quoted from [1]).**



**Figure 2: There should be a change in the rotation angle on the Z axis when users are in forward leaning posture.**

### 3.2 Implementing a posture improvement promoting system

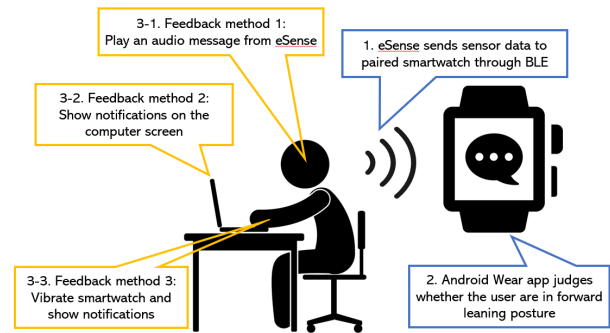
Since we can access the 6-axis accelerometer data through the BLE interface, we created an application to acquire the sensor values in real-time. To improve focus on the work or studies, it is recommended to keep smartphones out of sight. However, it is not the case for smartwatches, and smartwatches can also be used to collect further vital data that reflects user's state, such as hand motion and heart rate. Hence, we propose to collect eSense data from a BLE enable Android OS based smartwatch. Android API used to provide two ways for measuring the rotation angle of android devices, but `Sensor.TYPE_ORIENTATION` has already been deprecated, and `SensorManager.getOrientation()` cannot be used with eSense because it requires the value of a magnetometer, which eSense is not equipped. Therefore, we need to implement the algorithm to measure the rotation angle of the device by ourselves. We consider using the following two methods to measure the rotation angle and will verify their accuracy.

- Calculate the rotation angle by time integration of angular velocity, and correct it using the acceleration data.
- Calculate the rotation angle by applying a Kalman filter to acceleration data and angular velocity data.

Then, we propose three kinds of feedback methods to encourage users to improve their posture.

- The first method is to play an audio message from the eSense. It is inspired by the fact that people will wear earbuds while using this system, and it is expected to validate the effectiveness of auditory feedback while working.
- The second method is to show notifications on the computer screen. By using the OS notifications that users often see while working, it is expected to provide feedback without feeling users disturbed their work.
- The third method is to activate vibration and show notification on the smartwatch.

Figure 3 show a schematic of proposed system. We plan to evaluate the effectiveness of each method by asking the subjects to fill out a questionnaire and assess whether it helped improve their posture and whether it did not interfere with their work.



**Figure 3: Schematic overview of proposed posture improvement system**

## 4 FUTURE WORK

Both eSense and smartwatches do not interfere with work and do not cause discomfort even when worn for a long time. Besides measuring the forward-leaning posture from eSense, measuring the wrist tilt angle using the smartwatch and providing feedback to users may help prevent tendinitis caused by prolonged typing. Many smartwatches do not have a geomagnetic sensor but have an accelerometer and gyroscope. Hence, the method of measuring the rotation angle proposed in this study may be applicable.

In this way, we would like to develop a posture improvement promoting system that combines multiple daily usage devices and lead to the verification of its effectiveness.

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